

## Some of the key parameters used for evaluating LEDs

**Light output.** The light output of LEDs is normally the luminous flux, or energy per unit time that is radiated from a source over visible wavelengths. It refers mainly to energy radiated over wavelengths sensitive to the human eye, from about 330nm to 780nm. Flux is normally measured in lumen.

**Performance.** The performance of an LED is the light produced per diode and it depends on many factors ranging from material and size to temperature. Like the chip world has Moore's Law, the LED world has Haitz Law: "The amount of light that can be produced per diode (performance) would increase 20-fold every decade, while the cost of that light would decrease 10-fold."

While many generally use luminaire efficacy as a measure of performance, others believe that it should be seen as a standalone metric, but should be assessed in the light of other parameters like temperature, etc, relevant to the application. In general, high-performance LEDs offer a greater light output in a smaller footprint, and high luminous intensity for low currents.

**Efficiency and efficacy.** Similarly, both the efficacy and efficiency of LED bulbs have also been improving significantly. In lighting, efficacy refers to the amount of light (in lumens) produced by a certain amount of electricity (in watts). On the other hand, lighting fixture efficiency is the ratio of the total lumens exiting the fixture to the total lumens initially produced by the light source.

**Colour rendering index (CRI).** CRI is a measure of colour quality; that is, how well light sources render the colours of objects, materials, etc. The test procedure to arrive at the CRI number

involves comparing the appearance of eight colour samples under the light in question and a reference light source. The average differences measured are subtracted from 100 to get the CRI. So, small average differences will result in a higher score, while larger differences give a lower number. However, in recent times, the International Commission on Illumination (CIE) has raised some doubts as to the relevance of the older general colour rendering index (Ra) for measuring the quality of white LEDs.

To do away with any discrepancies, a long-term research and development process is underway to develop a revised colour quality metric that would be applicable to all white light sources. In the meantime, the CIE recommends that the CRI can be considered as one data point in evaluating white LED products and systems, but it should not be used to make product selections in the absence of in-person and on-site evaluations. The recommendations also add that if colour appearance is more important than colour fidelity, one should not exclude white light LEDs solely on the basis of relatively low CRI values. Some LED products with CRIs as low as 25 still produce visually pleasing white light.

**Lumen maintenance.** Lumen maintenance simply compares the amount of light produced from a light source or from a luminaire when it is brand new to the amount of light output at a specific time in the future.

In addition, the footprint, thermal management and optical control capabilities of LEDs are also considered, depending on the applications they are to be put to.

## Cost is going down too

According to some recent reports, improvements in device efficacy have been driving big improvements in lumens per dollar, but that is getting close to the theoretical maximum, and it is not easy to double it again. But, costs still need to come down further. So, the industry is turning to manufacturing efficiency for the solution to this problem.

Here, one of the main techniques to reduce cost is to use larger wafer substrates. Other areas of improvement include designing easy-to-manufacture devices, improving cost-of-ownership of equipment, better wafer-level testing and traceability for tracking defects back to root causes, better measurement of process conditions, improved ways to prepare the substrate before epitaxial film/wafer growth, improvement in the handling of the delicate wafers after laser lift-off, lower-cost ways to make patterned sapphire substrates, and so on.

"With latest manufacturing technologies, manufacturers are able to produce LEDs that are not only super bright but

also consume less power while producing light of higher intensity. Volume manufacturing on 15.24cm (6-inch) wafers started at the end of 2010, and massive transition from 5.08-10.16cm (2-inch to 4-inch) is set to begin in 2011. However, the adoption of 20.32cm (8-inch) wafers remains uncertain and will

## Uses aplenty

LEDs have innumerable uses, including...

1. Indicators on electronic devices, stereos, automobile dashboards, microwave ovens, etc.
2. Numeric displays on clocks, digital watches and calculators
3. Simple light bulbs for domestic and public lighting
4. Smart lighting systems for offices, homes
5. Short range optical signal transmission, such as in TV remote controls
6. Fashion accessories, such as blinking coloured lights on helmets
7. Back-lighting for liquid crystal display (LCD) screens
8. Automotive lighting
9. Power-smart displays for mobiles

require that sapphire wafer manufacture achieves significant cost reductions to make mass production more affordable," explains Chereddi.

Other technological developments like inorganic opto-semiconductors and organic light-emitting diodes with improved lumen efficiency and manufacturing processes might also reduce cost and enable mass manufacture in the near future.

And of course, there is the unbeatable force of the market. In India, the Bureau of Energy Efficiency (BEE) is coming up with a roadmap for enhancing the usage of priced LEDs in India. This would include standardising LEDs and developing testing, monitoring and verification protocols of LED lightings. The increasing applications and use by architects for aesthetic and energy-management applications will also increase demand and help reduce prices. Central government's 2011 budget has proposed to reduce excise duty and countervailing duty on imports of LED lights, which would further reduce the price of LED lightings in India.